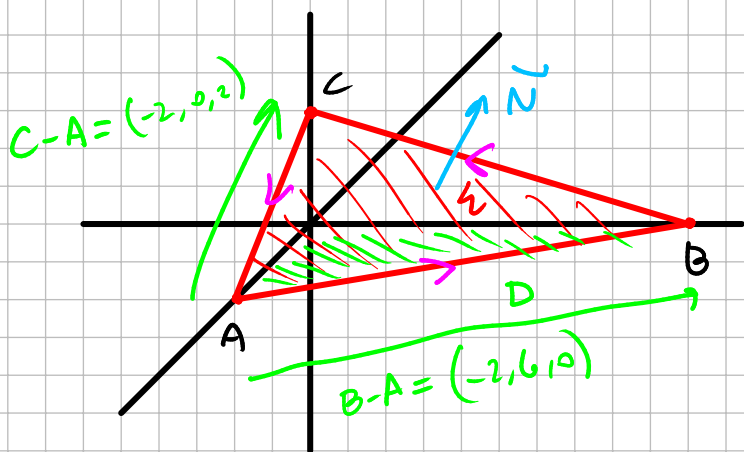


8. Calcular la integral de línea del campo vectorial dado por $\vec{F}(x, y, z) = (xz, -y, x^2y)$ a lo largo de la frontera del triángulo de vértices $A(2, 0, 0)$, $B(0, 6, 0)$ y $C(0, 0, 2)$ recorrida en este sentido.



CIRC
DEF
GREEN
F POT
STOKES

$$\nabla \times \vec{F} = (r'_y - a'_z, p'_z - r'_x, a'_x - p'_y)$$

$$\nabla \times \vec{F} = (x^2, x - 2xy, 0)$$

$$\oint_{\partial \Sigma} \vec{F} \cdot d\vec{l} = \iint_{\Sigma} \nabla \times \vec{F} \cdot d\vec{S} = \iint_D \nabla \times \vec{F}(\vec{\sigma}(u, v)) \cdot \vec{\sigma}'_u \times \vec{\sigma}'_v du dv$$

$$\vec{\sigma}(x, y) = (x, y, \dots)$$

$$(x, y, z) \cdot \vec{N} = \vec{N} \cdot \vec{P}$$

$$\vec{N} = \begin{pmatrix} -2, 0, 2 \\ -2, 6, 0 \end{pmatrix} \times \begin{pmatrix} -2, 6, 0 \end{pmatrix} = \begin{pmatrix} -12, -4, -12 \end{pmatrix}$$

$$(x, y, z) \cdot (-12, -4, -12) = (-12, -4, -12) \cdot (2, 0, 0)$$

$$-12x - 4y - 12z = -24$$

$$3x + y + 3z = 6$$

$$z = \frac{6 - 3x - y}{3}$$

$$\vec{\sigma}(x, y) = \left(x, y, \frac{6 - 3x - y}{3} \right)$$

$$\vec{\sigma}'_x = (1, 0, -1)$$

$$\vec{\sigma}'_y = \left(0, 1, -\frac{1}{3} \right)$$

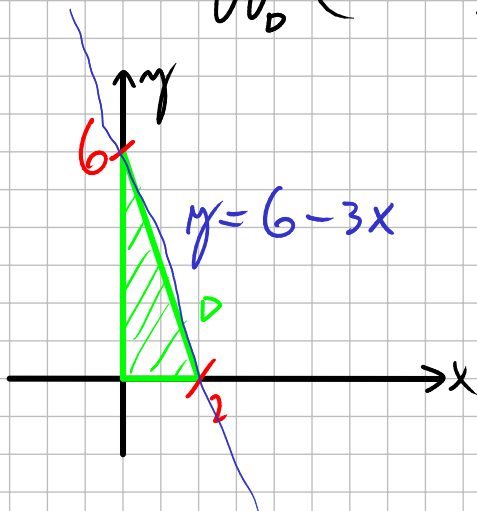
$$\vec{\sigma}'_x \times \vec{\sigma}'_y = \begin{pmatrix} 1, \frac{1}{3}, 1 \end{pmatrix} \checkmark$$

$$\iint_D (x^2, x - 2xy, 0) \cdot \left(1, \frac{1}{3}, 1 \right) dx dy$$

$$\iint_D (x^2, x - 2xy, 0) \cdot \left(1, \frac{1}{3}, 1\right) dx dy$$

$$\iint_D \left(x^2 + \frac{x}{3} - \frac{2}{3}xy\right) dx dy$$

$$\int_0^2 \int_0^{6-3x} \left(x^2 + \frac{x}{3} - \frac{2}{3}xy\right) dy dx = \frac{4}{3}$$



6. Sea $\vec{F} = (x^2, \cos(y) \sin(z), \sin(y) \cos(z))$.

(a) Demostrar que $\vec{F}$ es conservativo y encontrar una función potencial escalar f para $\vec{F}$

(b) Evaluar $\int_{\vec{x}} \vec{F} d\vec{s}$ a lo largo de la trayectoria $\vec{x}: [0, 1] \rightarrow \mathbb{R}^3$, $\vec{x}(t) = (t^2 + 1, e^t, e^{2t})$.

$$\vec{x}(0) = (1, 1, 1)$$

$$\vec{x}(1) = (2, e, e^2)$$

$$\nabla \times \vec{F} = (R'_y - Q'_z, P'_z - R'_x, Q'_x - P'_y)$$

$$= (\cos y \cdot \cos z - \cos y \cos z, 0, 0) = (0, 0, 0) \checkmark$$

$$\vec{F} \in C^1 \checkmark$$

$$\text{Dom } \vec{F}: \mathbb{R}^3 \text{ sc.}$$

$$\nabla \phi = \vec{F}$$

$$(\phi'_x, \phi'_y, \phi'_z) = (P, Q, R)$$

$$\int \phi'_x dx = \int x^2 dx$$

$$\phi = \frac{x^3}{3} + \alpha(y, z)$$

$$\phi'_y = \alpha'_y = \cos y \sin z$$

$$\int x' y dy = \int \cos y \sin z dy$$

$$\alpha(y, z) = \sin y \sin z + \beta(z)$$

$$\phi = \frac{x^3}{3} + \sin y \sin z + \beta(z)$$

$$\phi'_z = \sin y \cdot \cos z + \beta'(z) = \sin y \cdot \cos z$$

$$\beta'(z) = 0$$

$$\beta(z) = K$$

$$\phi = \frac{x^3}{3} + \sin y \sin z + K$$

$$\int_A^B \vec{F} \cdot d\vec{l} = \phi(B) - \phi(A)$$

$$= \phi(2, e, e^2) - \phi(1, 1, 1)$$

$$= \frac{8}{3} + \sin e \cdot \sin e^2 + K - \frac{1}{3} - \sin 1 \cdot \sin 1 - K$$

$$\frac{dy}{dx} - 2 \frac{y}{x} = x^2 \sin(3x)$$

$$y' + p(x)y = Q(x)$$

$$y' - \frac{2}{x}y = x^2 \sin(3x)$$

$$p(x) = -\frac{2}{x}$$

$$Q(x) = x^2 \sin(3x)$$

$$r(x) = e^{\int p(x) dx}$$

$$r'(x) = e^{\int p(x) dx} \cdot p(x) = r(x) \cdot p(x)$$

$$r'(x) = e^{\int p(x) dx} \cdot p(x) = r(x) \cdot p(x)$$

$$r(x)y' + r(x)p(x)y = Q(x) \cdot r(x)$$

$$r(x)y' + r'(x)y = Q(x) \cdot r(x)$$

$$\int (y \cdot r(x))' dx = \int Q(x) \cdot r(x) dx$$

$$y \cdot r(x) = \int Q(x) \cdot r(x) dx$$

$$y = \frac{\int Q(x) \cdot r(x) dx}{r(x)}$$

$$r(x) = e^{\int -\frac{2}{x} dx}$$

$$= e^{-2 \ln|x| + A}$$

$$= e^{\ln|x|^{-2}} \cdot e^A$$

$$= |x|^{-2} \cdot e^A$$

$$r(x) = \frac{e^A}{x^2} = \frac{B}{x^2}$$

$$y = \frac{\int x^2 \cdot \sin(3x) \cdot \frac{3}{x^2} dx}{3/x^2}$$

$$y = x^2 \int \sin(3x) dx = x^2 \left(\frac{-\cos(3x)}{3} + C \right)$$

$$y = -\frac{x^2}{3} \cdot \cos(3x) + C \cdot x^2$$

9. Sea V el sólido definido por $V = \{(x, y, z) \in \mathbb{R}^3 : x \geq 0, y \geq 0, z \geq 0, y + z \leq 4, x \leq 6\}$, sea S la superficie cerrada que limita a V . Calcular el flujo de salida a través de S del campo vectorial dado por $\vec{F}(x, y, z) = (xe^z, ye^z, e^z)$.

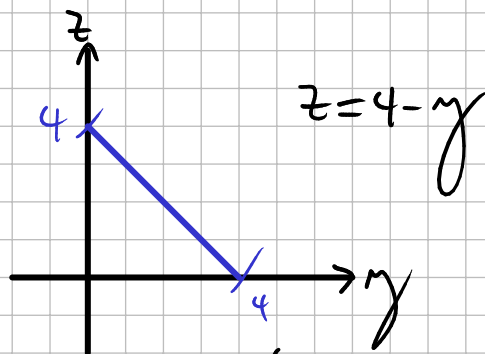
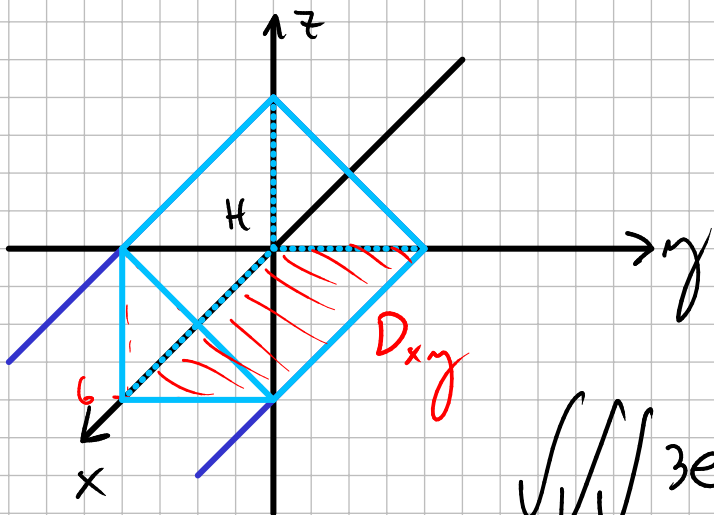
(a) ~~Directamente~~ **No se ja.**

(b) Mediante el teorema de Gauss.

$$\nabla \vec{F} = e^z + e^z + e^z = 3e^z$$

$$\oint_{\partial H} \vec{F} \cdot d\vec{S} = \iiint_H \nabla \cdot \vec{F} \cdot dV$$

$$V = \{(x, y, z) \in \mathbb{R}^3 : x \geq 0, y \geq 0, z \geq 0, y + z \leq 4, x \leq 6\},$$



$$\iiint_H 3e^z dV = \iint_{D_{xy}} \int_0^{4-y} 3e^z dz dx dy$$

$$= \iint_{D_{xy}} 3e^z \Big|_0^{4-y} dx dy = \iint_{D_{xy}} 3(e^{4-y} - 1) dx dy$$

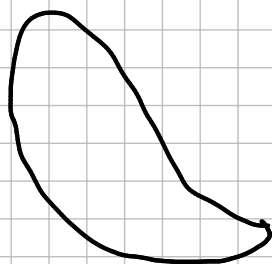
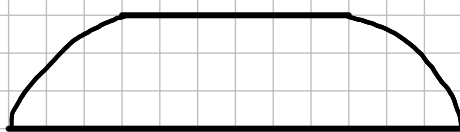
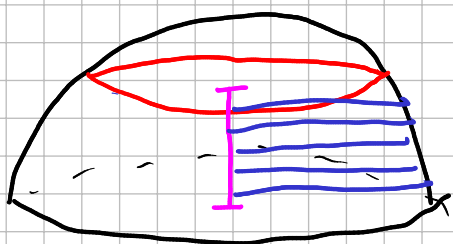
$$3 \iint_{D_{xy}} (e^{4-y} - 1) dx dy$$

$$3 \int_0^6 \int_0^4 (e^{4-y} - 1) dy dx$$

$$3 \int_0^6 \int_0^4 (e^{4-y} - 1) dy dx$$

$$3 \int_0^6 (-e^{4-y} - y) \Big|_0^4 dx$$

$$3 \int_0^6 (-1 - 4 + e^4) dx = 3 \cdot (e^4 - 5) \cdot x \Big|_0^6 = 18(e^4 - 5)$$



$$\vec{Q}(t) = \dots$$

$$\iint_D 1 dx dy$$

$$\oint_{\partial D^+} \vec{F} \cdot d\vec{l} = \iint_D \underbrace{(Q'_x - P'_y)}_1 dx dy$$

$$Q'_x - P'_y = 1$$

ELI5: $Q'_x = 1 \xrightarrow{\text{elijo}} Q = x$

$P'_y = 0 \longrightarrow P = 0$

$\rightarrow \vec{F} = (0, x)$

7. Evaluar $\oint_C (x^5 + 3y)dx + (2x - e^{y^3})dy$ donde C es la circunferencia dada por $(x-1)^2 + (y-5)^2 = 4$.

$$\int_C \vec{F} \cdot d\vec{l} = \int_C (P, Q) \cdot (dx, dy) = \int_C P dx + Q dy$$

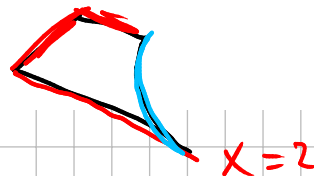
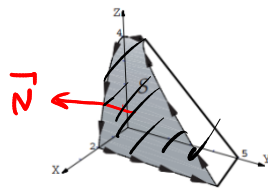
$$\vec{F} = (P, Q) = (x^5 + 3y, 2x - e^{y^3})$$

$$Q'_x - P'_y = 2 - 3 = -1$$

$$\oint_C \vec{F} \cdot d\vec{l} = \iint_D (Q'_x - P'_y) dx dy = \iint_D (-1) dx dy$$

$$= -\text{Area } D = -4\pi$$

7. Considerar el sólido Q limitado por la superficie $z = 4 - x^2$ y por el plano $y + z = 5$, en el primer octante. Calcular $\int_C \vec{F} \cdot d\vec{s}$ si $\vec{F}(x, y, z) = (yz, x, z^2)$ y C es la curva de la figura siguiente:



$$5 - y = z$$

$$5 - y = 4 - x^2$$

$$y = 1 + x^2$$

$$\nabla \times \vec{F} = (0, y, 1 - z)$$

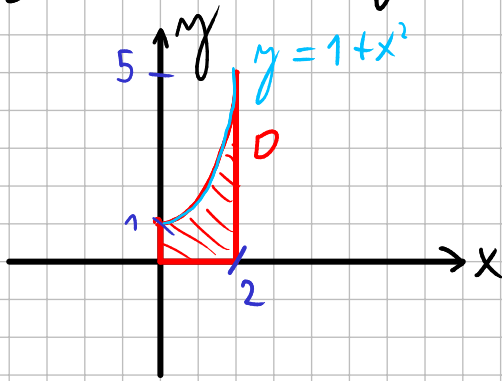
$$\oint_{\partial S} \vec{F} \cdot d\vec{l} = \iint_S \nabla \times \vec{F} \cdot d\vec{S} = \iint_D \nabla \times \vec{F}(\vec{\sigma}(u, v)) \cdot (\vec{\sigma}'_u \times \vec{\sigma}'_v) du dv$$

$$S: \vec{\sigma}(x, y) = (x, y, 4 - x^2) \quad (x, y) \in D$$

$$\begin{aligned} \vec{\sigma}'_x &= (1, 0, -2x) \\ \vec{\sigma}'_y &= (0, 1, 0) \\ \hline (2x, 0, 1) \end{aligned}$$

$$\iint_D (0, y, 1-4+x^2) \cdot (2x, 0, 1) dx dy$$

$$\iint_D (-3+x^2) dx dy = \int_0^2 \int_0^{1+x^2} (-3+x^2) dy dx$$



$$= \int_0^2 (-3+x^2)(1+x^2) dx$$

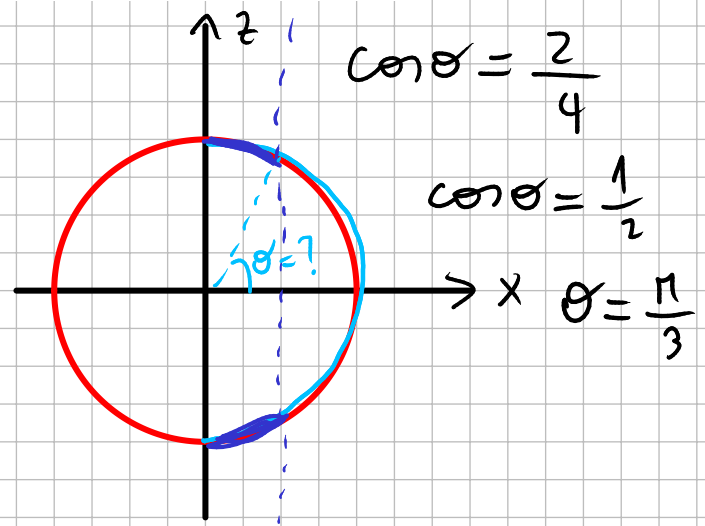
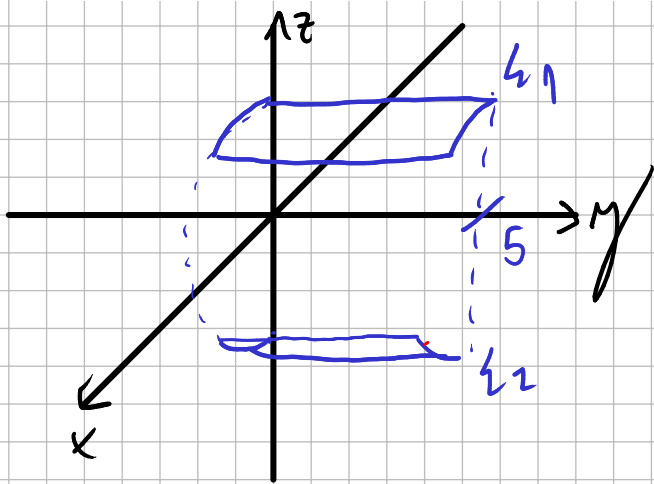
$$= \int_0^2 (-3-3x^2+x^2+x^4) dx$$

$$= \int_0^2 (-3-2x^2+x^4) dx$$

$$= \left(-3x - \frac{2}{3}x^3 + \frac{x^5}{5} \right) \Big|_0^2$$

$$= -\frac{74}{15}$$

7. Determinar el área de la superficie de la porción del cilindro $x^2 + z^2 = 16$ que está sobre la región en el primer cuadrante acotada por las gráficas de $x = 0$, $x = 2$, $y = 0$ e $y = 5$.



$$\Sigma = \xi_1 \cup \xi_2$$

$$\text{Area } \Sigma = 2 \cdot \text{Area } \xi_1$$

$$\text{Area } \xi_1 = \iint_{\xi_1} 1 \, dS = \iint_D \|\vec{\sigma}'_u \times \vec{\sigma}'_v\| \, du \, dv$$

$$\vec{\sigma}(\theta, y) = (4 \cos \theta, y, 4 \sin \theta)$$

$$\vec{\sigma}'_\theta = (-4 \sin \theta, 0, 4 \cos \theta)$$

$$\vec{\sigma}'_y = (0, 1, 0)$$

$$(-4 \cos \theta, 0, -4 \sin \theta)$$

$$(\theta, y) \in \left[\frac{\pi}{3}, \frac{\pi}{2}\right] \times [0, 5]$$

$$y \in [0, 5]$$

$$\theta \in \left[\frac{\pi}{3}, \frac{\pi}{2}\right]$$

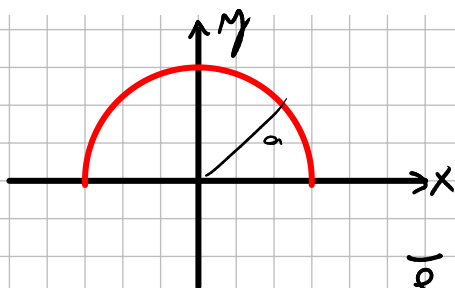
$$\|\vec{N}\| = \sqrt{16 \cos^2 \theta + 16 \sin^2 \theta} = 4$$

$$\int_0^5 \int_{\pi/3}^{\pi/2} 4 \, d\theta \, dy = 4 \cdot \frac{\pi}{6} \cdot 5 = \frac{10}{3} \pi$$

$$\text{Area } \Sigma = \frac{20}{3} \pi$$

8. Justificar que la masa total del alambre es $2ka^2$. Si este tiene la forma de una semicircunferencia de radio a y la densidad lineal de masa en un punto $P(x, y)$ es directamente proporcional a la distancia de P a la recta que pasa por los extremos del alambre. Siendo k la constante de proporcionalidad.

$$\text{Masa} = \int_C \delta \, dl$$



$$\delta = K \cdot |y| \stackrel{y \geq 0}{=} K \cdot y$$

$$\vec{\gamma}(t) = (a \cdot \cos t, a \cdot \sin t) \quad t \in [0, \pi]$$

$$\vec{\gamma}'(t) = (-a \sin t, a \cos t)$$

$$\|\vec{\gamma}'(t)\| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} = a$$

$$\int_C f \cdot dl = \int_{t_A}^{t_B} f(\vec{\gamma}(t)) \cdot \|\vec{\gamma}'(t)\| \, dt$$

$$= \int_0^\pi K \cdot a \cdot \sin t \cdot a \cdot dt = K \cdot a^2 \cdot 2$$

La curva de ecuación $\vec{X} = (t, t^k)$ con $t > 0$ es ortogonal a las curvas de la familia de ecuación $\frac{x^2}{4} + \frac{y^2}{9} = C$ con $C \in \mathbb{R}$ cuando:

$$y = x^k \rightarrow \ln y = \ln x^k \quad \begin{matrix} x=t \\ y=t^k \end{matrix}$$

$$y' = k \cdot x^{k-1}$$

$$\ln y = k \cdot \ln x$$

$$k = \frac{\ln y}{\ln x}$$

NoTaja.

$$\frac{x^2}{4} + \frac{y^2}{9} = C$$

$$y = f(x)$$

$$y' = f'(x)$$

$$(f^2(x))' = 2f(x) \cdot f'(x)$$

$$\frac{x}{2} + \frac{2y}{9} \cdot y' = 0$$

$$y' \Rightarrow -\frac{1}{y}$$

$$\frac{x}{2} - \frac{2y}{9} \cdot \frac{1}{y} = 0$$

$$\frac{x}{2} - \frac{2}{9} y \cdot \frac{dx}{dy} = 0$$

$$\frac{x}{2} = \frac{2}{9} y \frac{dx}{dy}$$

$$\int \frac{dy}{y} = \int \frac{4}{9} \frac{dx}{x}$$

$$\ln|y| = \frac{4}{9} \ln|x| + A$$

$$\ln|y| = \frac{4}{9} \ln|x| + A$$

$$|y| = e^{\frac{4}{9} \ln|x| + A}$$

$$|y| = e^A \cdot e^{\ln|x|^{\frac{4}{9}}}$$

$$|y| = e^A \cdot |x|^{\frac{4}{9}}$$

$$y = \pm e^A \cdot x^{\frac{4}{9}}$$

$$y = C \cdot x^{\frac{4}{9}}$$

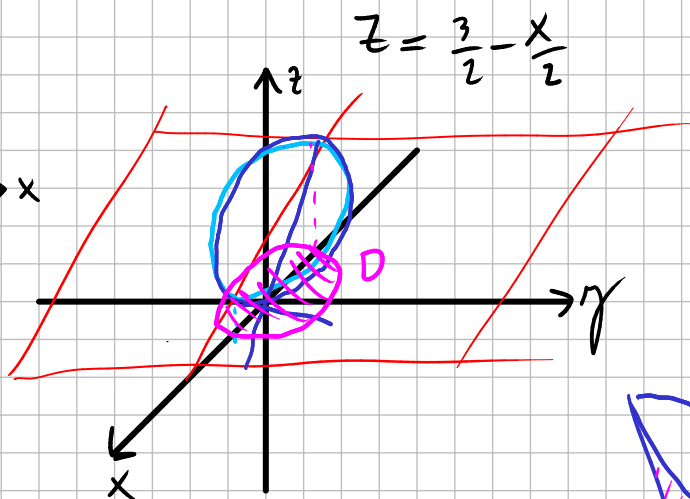
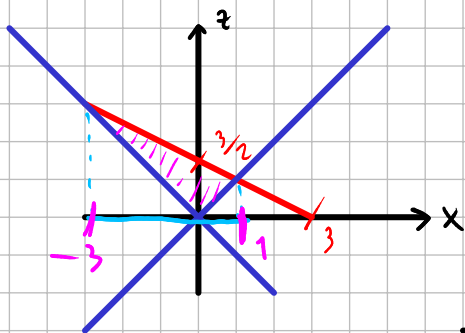
$$y = x^k$$

$$k = \frac{4}{9}$$

$$C = 1$$

$$y = x^{\frac{4}{9}}$$

4) Calcular el área de la porción del cono $z^2 = x^2 + y^2$ situada entre los planos $z = 0$ y $x + 2z = 3$.



$$\iint_S 1 ds = \iint_D \|\vec{\sigma}_u \times \vec{\sigma}_v\| du dv$$

$$\left(\frac{3}{2} - \frac{x}{2}\right)^2 = x^2 + y^2$$

$$\frac{1}{4}(3-x)^2 = x^2 + y^2$$

$$9 - 6x + x^2 = 4x^2 + 4y^2$$

$$9 = 3x^2 + 6x + 4y^2$$

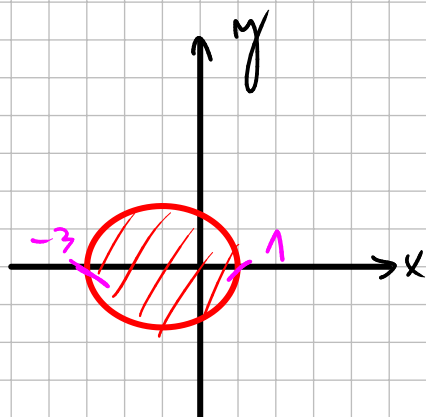
$$9 = 3(x^2 + 2x + 1 - 1) + 4y^2$$

$$9 = 3(x+1)^2 - 3 + 4y^2$$

$$9 = 3(x+1)^2 - 3 + 4y^2$$

$$12 = 3(x+1)^2 + 4y^2$$

$$1 = \frac{(x+1)^2}{4} + \frac{y^2}{3}$$



$$\vec{\sigma}(x, y) = (x, y, \sqrt{x^2 + y^2}) \quad (x, y) \in D$$

$$\vec{\sigma}'_x = (1, 0, \frac{x}{\sqrt{x^2 + y^2}})$$

$$\vec{\sigma}'_y = (0, 1, \frac{y}{\sqrt{x^2 + y^2}})$$

$$\left(-\frac{x}{\sqrt{x^2 + y^2}}, -\frac{y}{\sqrt{x^2 + y^2}}, 1 \right)$$

$$\|\vec{N}\| = \sqrt{\frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2} + 1} = \sqrt{2}$$

$$\text{Area } \Sigma = \iint_D \sqrt{2} \, dx \, dy = \sqrt{2} \cdot \text{Area } D = \sqrt{2} \cdot \pi \cdot 2 \cdot \sqrt{3} = 2\pi \cdot \sqrt{2} \cdot \sqrt{3}$$

Area ellipse

$$\int_0^1 \int_0^{2\pi} a b r \, d\theta \, dr = \pi \cdot a \cdot b$$

$$x = a \cos \theta + x_0$$

$$y = b \sin \theta + y_0$$

$$|J| = ab$$